

Leadless SMD Packages for C65FL Platform

Package Qualification Report

About this document

Scope and purpose

This document provides information about the results of the package qualification.

Intended audience

This Qualification Report is intended for customers of Infineon Technologies and certification parties. The data contained in this document is exclusively intended for technically trained staff.

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General Information

1 General Information

Table 1 General Information

Product Description	SOLID FLASH™ Controller for Security Applications
Wafer Process Technology	C65FL
Wafer Diameter	300 mm
Wafer Production Site	Taiwan Semiconductor Manufacturing Company, Ltd. (TSMC), Fab 14, Tainan, Taiwan R.O.C.
Preassembly Sites	Infineon Technologies AG, Regensburg, Germany Amkor Technology Philippines, Inc., Laguna, Philippines UTAC Thai Ltd., UTL3, Bang Pakong, Thailand
Assembly Sites	Amkor Technology Philippines, Inc., Laguna, Philippines UTAC Thai Ltd., UTL3, Bang Pakong, Thailand
Applied Qualification Standards	JESD47 ISO 7816 ISO 10373 ISO 14443
PCN / Info Note	2019-066-A0

Family Qualification with Generic Data from Reference Parts / Design / Products

2 Family Qualification with Generic Data from Reference Parts / Design / Products

Table 2 Family qualification for C65FL products with leadless SMD package technology Amkor

	Product	Wafer Technology	Wafer Fab	Package	Assembly Line
1. Part to be Qualified	16-bit Security Controller V02	C65FL	TSMC	PG-USON-10-2	Amkor
2. Part to be Qualified	16-bit Security Controller V01	C65FL	TSMC	PG-USON-10-2	Amkor
3. Part to be Qualified	16-bit Security Controller V03	C65FL	TSMC	PG-USON-10-2	Amkor
4. Part to be Qualified	32-bit ARM based Security Controller V07	C65FL	TSMC	PG-UQFN-32-1	Amkor
5. Part to be Qualified	16-bit Security Controller V05	C65FL	TSMC	PG-UQFN-32-1	Amkor
6. Part to be Qualified	16-bit Security Controller V03	C65FL	TSMC	PG-VQFN-32-13	Amkor
7. Part to be Qualified	32-bit ARM based Security Controller V07	C65FL	TSMC	PG-VQFN-32-13	Amkor
8. Part to be Qualified	32-bit ARM based Security Controller V01	C65FL	TSMC	PG-VQFN-8-4	Amkor
9. Part to be Qualified	16-bit Security Controller V07	C65FL	TSMC	PG-USON-10-2	Amkor
10. Part to be Qualified	16-bit Security Controller V08	C65FL	TSMC	PG-USON-10-2	Amkor
Reference Part SD	M5072G	L90FL	TSMC	PG-VQFN-32-13	Amkor

Family Qualification with Generic Data from Reference Parts / Design / Products

Table 3 Family qualification for C65FL products with leadless SMD package technology UTAC

	Product	Wafer Technology	Wafer Fab	Package	Assembly Line
1. Part to be Qualified	32-bit ARM based Security Controller V07	C65FL	TSMC	PG-VQFN-32-20	UTAC
2. Part to be Qualified	16-bit Security Controller V03	C65FL	TSMC	PG-VQFN-32-20	UTAC
3. Part to be Qualified	16-bit Security Controller V01	C65FL	TSMC	PG-USON-10-4	UTAC
4. Part to be Qualified	16-bit Security Controller V02	C65FL	TSMC	PG-USON-10-4	UTAC
Reference Part A	16-bit Security Controller V01	C65FL	TSMC	PG-USON-10-2	Amkor
Reference Part B	16-bit Security Controller V02	C65FL	TSMC	PG-USON-10-2	Amkor
Reference Part SD	SLI97	L90FL	Infineon	PG-VQFN-32-20	UTAC

Qualification Results

3 Qualification Results

3.1 Qualification Results for Amkor PG-USON-10-2 & PG-UQFN-32-1

Table 4 Package Qualification Result Overview

Qualification Test	Test Conditions / Reference / Acceptance Criteria ¹	# lots x min. SS per Lot	1. Part to be Qualified 16-bit Security Controller V02 with PG-USON-10-2	2. Part to be Qualified 16-bit Security Controller V01 with PG-USON-10-2	3. Part to be Qualified 16-bit Security Controller V03 with PG-USON-10-2	4. Part to be Qualified 32-bit ARM based Security Controller V07 with PG-UQFN-32-1	5. Part to be Qualified 16-bit Security Controller V05 with PG-UQFN-32-1	9. Part to be Qualified 16-bit Security Controller V07 with PG-USON-10-2	10. Part to be Qualified 16-bit Security Controller V08 with PG-USON-10-2	Reference Part SD M5072G with PG-VQFN-32-13
			Ref. 1	Ref. 2	Ref. 3	Ref. 4	Ref. 5	Ref. 9	Ref. 10	Ref. SD
Preconditioning	MSL1, 3x Reflow @ 260 °C JESD22-A113 EIA/J-STD-020	All samples prior TC, THB, UHAST	PASS	PASS	PASS ²	PASS	PASS	PASS	PASS and Ref. 1	---
High Temperature Storage	150 °C, 1000 h JESD22-A103	3 x 25	PASS	Ref. 1	Ref. 1	PASS	Ref. 4	Ref. 4	Ref. 1	---
Temperature Cycling	-40 °C ... +115 °C, 1000 cycles JESD22-A104, Cond. I	3 x 25	PASS	PASS	Ref. 1,5,6	PASS	PASS	PASS	PASS and Ref. 1	---
Temperature Humidity Bias Test	85 °C, 85 % r.h., 5.5V, 1000 h JESD22-A101	3 x 25	PASS	Ref. 1	Ref. 1	PASS	Ref. 4	Ref. 4	Ref. 1	---
Unbiased Highly Accelerated Stress Test	130 °C, 85 % r.h., 96 h JESD22-A118	3 x 25	PASS	Ref. 1	Ref. 1	PASS	Ref. 4	Ref. 4	Ref. 1	---
Electrostatic Discharge Charged Device Model	750 V, 3 devices per voltage class JS-002-2014	1 x 3	PASS	PASS	PASS	PASS	PASS	PASS	PASS	---
Bond Pull Strength	30 bonds / 5 units M2011	3 x 10	PASS	PASS	PASS	PASS	PASS	PASS	PASS	---
Bond Shear	30 bonds / 5 units JESD22-B116	3 x 10	PASS	PASS	PASS	PASS	PASS	PASS	PASS	---
Package Dimensions	Package outline	3 x 10	PASS	PASS	PASS	PASS	PASS	PASS	PASS	---
Solderability	Aging, wetting, dewetting J-STD-002D IEC 60068-2-20-20	3 x 22	Ref. SD							PASS

¹ Acceptance Criteria is "No fails" in case nothing else is specified

² Preconditioning was done with 3 x 25 samples. No fails observed.

Qualification Results

3.2 Qualification Results for Amkor PG-VQFN-32-13 & PG-VQFN-8-4

Table 5 Package Qualification Result Overview

Qualification Test	Test Conditions / Reference / Acceptance Criteria ¹	# lots x min. SS per Lot	6. Part to be Qualified: 16-bit Security Controller V03 with PG-VQFN-32-13	7. Part to be Qualified: 32-bit ARM based Security Controller V07 with PG-VQFN-32-13	8. Part to be Qualified: SLC37M with PG-VQFN-8-4	Reference Part SD: M5072G with PG-VQFN-32-13
			Ref. 6	Ref. 7	Ref. 8	Ref. SD
Preconditioning	MSL1, 3x Reflow @ 260 °C JESD22-A113 EIA/J-STD-020	All samples prior TC, THB, UHA8T	PASS	PASS	PASS	---
High Temperature Storage	150 °C, 1000 h JESD22-A103	3 x 25	PASS	Ref. 6	Ref. 6	---
Temperature Cycling	-40 °C ... +115 °C, 1000 cycles JESD22-A104, Cond. I	3 x 25	PASS	PASS	PASS	---
Temperature Humidity Bias Test	85 °C, 85 % r.h., 5.5V, 1000 h JESD22-A101	3 x 25	PASS	Ref. 6	Ref. 6	---
Unbiased Highly Accelerated Stress Test	130 °C, 85 % r.h., 96 h JESD22-A118	3 x 25	PASS	Ref. 6	Ref. 6	---
Electrostatic Discharge Charged Device Model	750 V, 3 devices per voltage class JS-002-2014	1 x 3	PASS	PASS	PASS	---
Bond Pull Strength	30 bonds / 5 units M2011	3 x 10	PASS	PASS	PASS	---
Bond Shear	30 bonds / 5 units JESD22-B116	3 x 10	PASS	PASS	PASS	---
Package Dimensions	Package outline	3 x 10	PASS	PASS	PASS	---
Solderability	Aging, wetting, dewetting J-STD-002D IEC 60068-2-20-20	3 x 22	Ref. SD			PASS

¹ Acceptance Criteria is "No fails" in case nothing else is specified

Qualification Results

3.3 Qualification Results for UTAC PG-VQFN-32-20 & PG-USON-10-4

Table 6 Package Qualification Result Overview

Qualification Test	Test Conditions / Reference / Acceptance Criteria ¹	# lots / min. sample size per lot	1. Part to be Qualified: 32-bit ARM based Security Controller V07 with PG-VQFN-32-20	2. Part to be Qualified: 16-bit Security Controller V03 with PG-VQFN-32-20	3. Part to be Qualified: 16-bit Security Controller V01 with PG-USON-10-4	4. Part to be Qualified: 16-bit Security Controller V02 with PG-USON-10-4	Reference part A: 16-bit Security Controller V01 with PG-USON-10-2	Reference part B: 16-bit Security Controller V02 with PG-USON-10-2	Reference Part SD: SLI 97 With PG-VQFN-32-20
			Ref 1	Ref 2	Ref 3	Ref 4	Ref A	Ref B	Ref SD
Charged Device Model ESD (ESD CDM)	750 V, 25 °C JS-002-2014	1 x 3	PASS	PASS	Ref A	Ref B	PASS	PASS	---
High Temperature Storage Life (HTSL)	150 °C, 1000 h, condition A JESD22-A103	3 x 25	PASS	Ref 1	PASS	Ref 3	---	---	---
Preconditioning (PC)	MSL 1, peak temp.: 260°C JESD22-A113	All samples prior TC, THB, uHAST	PASS	PASS	PASS	PASS	---	---	---
Temperature Humidity Bias (THB)	85°C, 85% RH, 5.5V, 1000h, JESD22-A101	3 x 25	PASS	Ref 1	PASS	Ref 3	---	---	---
Temperature Cycling (TC)	-55/+125°C, 700x, condition B, JESD22-A104	3 x 25	PASS	PASS	PASS	PASS ² Ref 3	---	---	---
Unbiased Highly Accelerated Stress Test (UHAST)	130°C, 85% RH, 230kPa, 96h, JESD22-A118	3 x 25	PASS	Ref 1	PASS	Ref 3	---	---	---
Bond Pull Strength (BPS)	2.3 cN, 5 bonds / unit, MIL-STD-883, Method 2011	1 x 6	PASS	PASS	PASS	PASS	---	---	---
Bond Shear (BS)	13 cN, 5 bonds / unit, JESD22-B116	1 x 6	PASS	PASS	PASS	PASS	---	---	---
Package Dimensions	Package outline	1 x 30	PASS	PASS	PASS	PASS			
Solderability (SD)	Aging, wetting, dewetting J-STD-002D	1 x 15	Ref SD	Ref SD	Ref 4	PASS	---	---	PASS

¹ Acceptance Criteria is "No fails" in case nothing else is specified

² 1 lot x 77 samples

Qualification Results

Revision History

Major changes since the last revision

Page or Reference	Description of change
Revision 1.2, 2020-02-26	
4, 6	General review
5, 8	New product qualified: 16-bit Security Controller V08
	New packages at assembly site UTAC: PG-VQFN-32-20 and PG-USON-10-4
Revision 1.1, 2018-09-12	
2, 3	New product qualified: 16-bit Security Controller V07 with PG-USON-10-2
	Typo corrected in Table 3.1
Revision 1.0, 2018-07-09	
All	First issue

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